Embedded Systems Interface Analysis Report

System 1: Microwave Oven

System Identification

Its primary purpose is to heat or cook food using microwave radiation. A built-in microcontroller manages cooking time, power level, safety mechanisms, and user interaction.

Input and Output Description

Inputs:

- Keypad or touch buttons for selecting cooking options
- Door switch sensor to detect whether the door is open or closed
- Rotary dial (on some models) to adjust time or power
- Internal temperature or humidity sensors (on advanced models)

Outputs:

- LCD or LED display showing cooking time and settings
- Interior light
- Beep or buzzer when cooking is complete
- Magnetron activation to generate microwaves
- Rotating turntable motor

Interface Classification

Interface	Type	Classification
LCD/LED Display	Output	Visual
Interior Light	Output	Visual
Keypad/Touch Buttons	Input	Tactile
Rotary Dial	Input	Tactile
Door Switch	Input	Tactile
Beeper/Buzzer	Output	Audio

Temperature/Humidity Sensor	Input	Electronic
Interface	**Type**	**Classification**
Magnetron Control	Output	Electronic
Turntable Motor Control	Output	Electronic

System 2: Smart Thermostat (Google Nest)

System Identification

Google Nest Smart Thermostat's primary purpose is to automatically regulate indoor temperature while improving energy efficiency. It uses embedded processing to monitor environmental conditions and control heating and cooling systems.

Input and Output Description

Inputs:

- Touch display
- Temperature sensor
- Motion (occupancy) sensor
- Humidity sensor
- Smartphone application commands
- Wi-Fi network communication

Outputs:

- LCD display
- HVAC control signals
- Mobile notifications
- Status LEDs
- Wireless communication with cloud services

Interface Classification

Interface	Type	Classification
LCD Display	Output	Visual
Status LEDs	Output	Visual

Touchscreen	Input	Tactile
Interface	**Type**	**Classification**
Temperature Sensor	Input	Electronic
Humidity Sensor	Input	Electronic
Motion Sensor	Input	Electronic
HVAC Control Signals	Output	Electronic
Wi-Fi Communication	Input/Output	Electronic
Smartphone Commands	Input	Electronic
Mobile Notifications	Output	Electronic

Conclusion

Both the microwave oven and the Google Nest Smart Thermostat are examples of embedded systems designed for specific tasks. The microwave primarily relies on **tactile, visual, audio, and internal electronic interfaces** to provide a simple user experience. In contrast, the smart thermostat extends traditional embedded functionality by incorporating **electronic communication through Wi-Fi**, allowing remote monitoring and automation via a smartphone application. While both systems process user inputs and control hardware outputs, the thermostat demonstrates how modern embedded systems increasingly depend on network connectivity and smart automation to improve convenience, energy efficiency, and user control.